THE STACK OF COHERENT SHEAVES IS FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1132. Standing situation.
- Definition categories-definition-fibred-groupoids (canonical tag 003T). cited at proof line 1140. The conditions.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at statement line: the lemma is phrased in this situation; load-bearing.

Source reference. Original definition 003T: sources/stacks-chow/tags.txt, source lines 147–147. Cited at proof line 1140: the two conditions.

Lemma 0.2. *In Situation 08KB the functor $p : \mathcal{C}oh_{X/B} \longrightarrow (Sch/S)_{fppf}$ is fibred in groupoids.*

Proof. We show that p is fibred in groupoids by checking conditions (1) and (2) of Categories, Definition 003T. Given an object $(T', g', \mathcal{F}')$ of $\mathcal{C}oh_{X/B}$ and a morphism $h : T \to T'$ of schemes over S we can set $g = h \circ g'$ and $\mathcal{F} = (h')^*\mathcal{F}'$ where $h' : X_T \to X_{T'}$ is the base change of h. Then it is clear that we obtain a morphism $(T, g, \mathcal{F}) \to (T', g', \mathcal{F}')$ of $\mathcal{C}oh_{X/B}$ lying over h. This proves (1). For (2) suppose we are given morphisms

$$(h_1, \varphi_1) : (T_1, g_1, \mathcal{F}_1) \to (T, g, \mathcal{F}) \quad \text{and} \quad (h_2, \varphi_2) : (T_2, g_2, \mathcal{F}_2) \to (T, g, \mathcal{F})$$

of $\mathcal{C}oh_{X/B}$ and a morphism $h : T_1 \to T_2$ such that $h_2 \circ h = h_1$. Then we can let φ be the composition

$$(h')^*\mathcal{F}_2 \xrightarrow{(h')^*\varphi_2^{-1}} (h')^*(h_2)^*\mathcal{F} = (h_1)^*\mathcal{F} \xrightarrow{\varphi_1} \mathcal{F}_1$$

to obtain the morphism $(h, \varphi) : (T_1, g_1, \mathcal{F}_1) \to (T_2, g_2, \mathcal{F}_2)$ that witnesses the truth of condition (2). □

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